Sales Data Analysis Using SQL

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Total revenue

- select sum(quantity*price)as Total_Revenue from sales;

	Total_Revenue
▶	56650.00

Total orders

- select count(order_id) as total_orders from sales;

	total_orders
▶	15

Monthly sales trend

- select month(order_date) as month, sum(quantity*price) as revenue from sales group by month order by month;

	month	revenue
▶	1	35950.00
	2	20700.00

“ Sales show variation across months, indicating possible seasonality ”

Revenue by product category

- select product_category, sum(quantity*price) as revenue from sales group by product_category order by revenue desc;

	product_category	revenue
▶	Electronics	16000.00
	Grocery	13900.00
	Home	11000.00
	Clothing	9600.00
	Beauty	6150.00

Running Revenue

- select order_date, sum(quantity*price) as daily_revenue, sum(sum(quantity*price)) over (order by order_date) as running_total from sales group by order_date;

order_date	daily_revenue	running_total
2024-01-05	5000.00	5000.00
2024-01-06	3200.00	8200.00
2024-01-07	6000.00	14200.00
2024-01-08	2550.00	16750.00
2024-01-09	5500.00	22250.00
2024-01-10	2800.00	25050.00
2024-01-11	4200.00	29250.00
2024-01-12	1900.00	31150.00
2024-01-13	3600.00	34750.00
2024-01-14	1200.00	35950.00
2024-02-01	1500.00	37450.00
2024-02-03	6800.00	44250.00
2024-02-05	6000.00	50250.00
2024-02-07	2400.00	52650.00
2024-02-10	4000.00	56650.00

City Performance Analysis

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select city, sum(quantity*price) as revenue ,rank() over(order by
sum(quantity*price) desc) as City_rank from sales group by city;
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city	revenue	City_rank
Lucknow	12000.00	1
Mumbai	10000.00	2
Chennai	9500.00	3
Delhi	6500.00	4
Ahmedabad	4950.00	5
Jaipur	4200.00	6
Surat	3600.00	7
Bangalore	2800.00	8
Pune	1900.00	9
Kochi	1200.00	10

"Top-Ranked Cities contribute the highest revenue, under performing cities may need target strategies"

Customer Type

- select customer_name, count(order_id) as Total_orders ,case when count(order_id)=1 then "New" else "Repeated_Customer" end as Customer_type from sales group by customer_name;

customer_name	Total_orders	Customer_type
Amit Sharma	2	Repeated_Customer
Neha Verma	2	Repeated_Customer
Rahul Singh	2	Repeated_Customer
Priya Mehta	2	Repeated_Customer
Suresh Kumar	2	Repeated_Customer
Anjali Rao	1	New
Vikas Gupta	1	New
Kavita Joshi	1	New
Rohit Patel	1	New
Sneha Iyer	1	New

" A majority of customers are repeat buyers, indicating strong customer retention"

Top customers by spending

- select customer_name, sum(price*quantity) as total_spent from sales group by customer_name order by total_spent desc limit 5;

	customer_name	total_spent
▶	Rahul Singh	12000.00
	Neha Verma	10000.00
	Suresh Kumar	9500.00
	Amit Sharma	6500.00
	Priya Mehta	4950.00

Top 2 Selling Product

- select product_category, sum(quantity) as total_quantity from sales group by product_category order by total_quantity desc limit 2 ;

product_category	total_quantity
Grocery	17
Clothing	7

Under Performing Cities by Revenue

- select city, sum(quantity*price) as revenue from sales group by city order by revenue ASC limit 2;

	city	revenue
▶	Kochi	1200.00
	Pune	1900.00

Orders by status

- select order_status, count(*) as total_orders from sales group by order_status;

	order_status	total_orders
▶	Delivered	13
	Cancelled	1
	Returned	1

Cancelled / Returned Orders

- select * from sales where order_status in ("cancelled","returned");

order_id	order_date	customer_id	customer_name	city	product_category	quantity	price	order_status
4	2024-01-08	4	Priya Mehta	Ahmedabad	Beauty	3	850.00	Cancelled
7	2024-01-11	7	Vikas Gupta	Jaipur	Home	1	4200.00	Returned
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Count

select count(*) as Cancelled_Returned_count from sales where order_status in ("cancelled","returned");

Cancelled_Returned_count
2

Average Order Value by City

- select city, avg(quantity*price) **as** avg_order_value from sales group by city order by avg_order_value desc ;

	city	avg_order_value
▶	Lucknow	6000.000000
	Mumbai	5000.000000
	Chennai	4750.000000
	Jaipur	4200.000000
	Surat	3600.000000
	Delhi	3250.000000
	Bangalore	2800.000000
	Ahmedabad	2475.000000
	Pune	1900.000000
	Kochi	1200.000000

Payment Status Of all Customers (Left join)

- Select sales.order_id,sales.customer_name,payments.payment_method, payments.payment_status from sales left join payments on sales.order_id=payments.order_id;

order_id	customer_name	payment_method	payment_status
1	Amit Sharma	Credit card	Completed
2	Neha Verma	UPI	Completed
3	Rahul Singh	Cash	Completed
4	Priya Mehta	Debit card	Completed
5	Suresh Kumar	Credit card	Pending
6	Anjali Rao	UPI	Failed
7	Vikas Gupta	Cash	Completed
8	Kavita Joshi	Debit card	Pending
9	Rohit Patel	Credit card	Completed
10	Sneha Iyer	UPI	Pendind
11	Amit Sharma	Cash	Completed
12	Neha Verma	Debit card	Completed

Failed Payment Analysis (Right join)

- select payments.order_id,sales.customer_name,sales.price ,payments.payment_method,payments.payment_status from payments right join sales on sales.order_id=payments.order_id where payment_status="Failed";

	order_id	customer_name	price	payment_method	payment_status
▶	6	Anjali Rao	700.00	UPI	Failed

“Failed payments may indicate transaction issue or system inefficiencies”

Total Payments

- select payments.payment_status,count(*) as payment_status from payments group by payment_status;

	payment_status	payment_status
▶	Completed	11
	Pending	3
	Failed	1
	Pendind	1

Full Join

Select sales.order_id,sales.customer_name,payments.payment_method,

payments.payment_status from sales left join payments on sales.order_id=payments.order_id union select sales.order_id, sales.customer_name,payments.payment_method,payments.payment_status from sales right join payments on sales.order_id=payments.order_id

13	Rahul Singh	Credit card	Completed
14	Priya Mehta	UPI	Completed
15	Suresh Kumar	Cash	Completed
NULL	NULL	Debit card	Pending

Conclusion

- Revenue varies across months, indicating seasonal trends
- A small number of cities contribute a large share of revenue
- High repeat customers show good retention
- Failed payments highlight areas for operational improvement

"THANK YOU"